Fleet Modem Diagnostic Report

Sierra Wireless EM7565 — Machines N120AV & N096AV

Date: June 18, 2026 | Prepared by: Field Diagnostics

Machine	Modem Serial	Issue	Verdict
N120AV	UF21143469021050	SIM not detected after interrupted firmware flash	Replace modem
N096AV	UF22173591021052	NO IMSI — QMI/UIM stack failure (firmware upgraded, persists)	Replace modem

Both modems have been diagnosed as having permanent hardware defects in their SIM interface controllers. Modem replacement is required for both machines. This report documents the full diagnostic process, evidence gathered, and actions taken for each machine.

Report 1 of 2: Machine N120AV

Modem: Sierra Wireless EM7565 | Serial: UF21143469021050

1. Executive Summary

Machine N120AV experienced a complete cellular connectivity failure. After systematic diagnosis, the root cause was determined to be a damaged Sierra Wireless EM7565 modem. The modem is unable to detect the SIM card, preventing network registration and PPP data session establishment. A modem replacement is required.

2. System Environment

Component	Details
Machine ID	N120AV
OS	Ubuntu 20.04 LTS
Modem	Sierra Wireless EM7565 (Qualcomm Snapdragon X16 LTE-A)
Modem Serial	UF21143469021050
Firmware (post-upgrade)	SWI9X50C_01.14.02.00 (August 2020)
USB Mode	MBIM (cdc_mbim) — incorrect; should be QMI
SIM Provider	Movistar Colombia (MCC/MNC: 732123)
Connectivity	PPP via pppd + APN: internet.movistar.com.co
Remote Access	OpenVPN over PPP

3. Initial Symptom

The PPP connection established LCP and PAP authentication successfully but timed out waiting for IPCP (IP address assignment). The ppp0 interface showed heavy TX traffic but near-zero RX bytes frozen at 62.

```
PAP authentication succeeded
IPCP: timeout sending Config-Requests
Connection terminated. Modem hangup
```

4. Diagnostic Steps Performed

Step 1 — PPP Traffic Analysis

Interface statistics confirmed outbound packets sent but nothing received:

```
RX: bytes=62 packets=5
TX: bytes=145399 packets=2194
```

Step 2 — ModemManager vs pppd Conflict Check

Compared ModemManager status between the working reference machine (N133AV) and N120AV. N133AV had ModemManager managing the modem via QMI with qmi-proxy running. N120AV had ModemManager failing to initialize the modem entirely.

Step 3 — USB Composition Mode Identified

Interface	N120AV (Broken)	N133AV (Working)	Impact
If#0	cdc_mbim (MBIM)	qmi_wwan (QMI)	Wrong protocol mode
If#2	qcserial (AT)	qcserial (AT)	OK
If#3	qcserial (PPP)	qcserial (PPP)	OK

The modem was in MBIM composition mode instead of QMI, preventing ModemManager from initializing it correctly.

Step 4 — Attempted USB Composition Switch

Attempted to switch modem from MBIM to QMI mode via AT commands:

```
AT!ENTERCND="A710" -> OK
AT!USBCOMP? -> 0000100D (diag,nmea,modem,mbim)
AT!USBCOMP=1,1,0000500D -> OK
AT!RESET
```

After reset the modem came back still in MBIM mode — the composition change did not persist.

Step 5 — Interrupted Firmware Upgrade

A firmware upgrade was attempted but interrupted mid-process with Ctrl+C. After reboot:

- AT+CIMI returned ERROR — cannot read SIM IMSI
- AT+CSQ returned ERROR — cannot read signal strength
- AT+CREG? returned ERROR — cannot access network registration
- AT!GSTATUS? showed: MM (CS) state: NO IMSI, PS state: Not attached

Step 6 — Post-Reimage Assessment

A full OS reimage was performed. After reimaging:

- Modem still in MBIM mode (USB composition stored in modem NV memory, not OS)
- AT command ports (ttyUSB3, ttyUSB4) returned no response
- ModemManager: No modems were found

Step 7 — MBIM Interface Communication

Installed libmbim-utils and communicated via /dev/cdc-wdm0:

MBIM Query	Result
--query-device-caps	Firmware: SWI9X50C_01.14.02.00 (upgraded successfully)
--query-subscriber-ready-status	Ready state: sim-not-inserted
--query-signal-state	RSSI: 99 (no signal)
--query-registration-state	Register state: searching (no SIM to register)
--set-radio-state=off/on	No change — SIM still not detected after radio cycle

5. Conclusion

The diagnostic evidence points conclusively to a damaged Sierra Wireless EM7565 modem:

- Firmware upgrade completed successfully (SWI9X50C_01.14.02.00 confirmed via MBIM).
- SIM card is functional — tested working in other machines.
- SIM interface non-functional: sim-not-inserted reported even with known-good SIM present.
- AT command ports completely unresponsive despite no process holding them.
- Radio power cycling via MBIM did not restore SIM detection.
- OS reimage had no effect — fault is in modem hardware/NV memory.

Most likely cause: the interrupted firmware upgrade corrupted the modem's NV memory, permanently damaging the SIM interface controller. This is not recoverable through software means.

6. Recommendations

Priority	Action	Details
Immediate	Replace modem	Replace Sierra Wireless EM7565 on N120AV. Not recoverable.
Before replacement	Verify QMI mode	Confirm replacement modem enumerates with qmi_wwan driver.
Fleet-wide	Audit USB composition	Run: usb-devices | grep -A10 sierra on all machines.
Preventive	Never interrupt firmware flash	Allow flash to complete or perform full recovery flash.

Report 2 of 2: Machine N096AV

Modem: Sierra Wireless EM7565 | Serial: UF22173591021052 | Version 2.0

1. Executive Summary

Machine N096AV has no cellular connectivity. After extensive diagnosis including a successful firmware upgrade from SWI9X50C_01.08.04.00 to SWI9X50C_01.14.02.00, the modem continues to report NO IMSI internally despite the SIM card being physically readable and signal being strong (CSQ 25/31). All software-level recovery options have been exhausted. The modem hardware is defective and must be replaced.

2. System Environment

Component	Details
Machine ID	N096AV
OS	Ubuntu 20.04 LTS
Modem	Sierra Wireless EM7565 (Qualcomm Snapdragon X16 LTE-A)
Modem Serial	UF22173591021052
Original Firmware	SWI9X50C_01.08.04.00 (August 2018)
Updated Firmware	SWI9X50C_01.14.02.00 (August 2020)
USB Mode	QMI (qmi_wwan)
SIM Provider	Movistar Colombia (MCC/MNC: 732123)
SIM IMSI	732123667007142
SIM ICCID	8957123412113518379F
Location	Same physical location as working machines N133AV and N151AV

3. Initial Symptom

PPP connection established LCP and PAP authentication successfully but timed out at IPCP. RX bytes frozen at 62 despite active TX traffic.

```
PAP authentication succeeded
IPCP: timeout sending Config-Requests
Connection terminated. Modem hangup
```

4. Diagnostic Steps Performed

Step 1 — ModemManager Disabled State

ModemManager detected the modem via QMI but reported it as disabled at boot. Fixed by deploying modem-enable.service systemd unit (fleet-wide fix also applied to N133AV and N151AV).

Step 2 — ttyUSB Port Mapping Mismatch

Symlink	Points To	Actual Function (per MM)
modemAT	ttyUSB4	GPS port — AT commands not accepted
modemPPP	ttyUSB5	Actual AT interface

Step 3 — Operator Registration Drift

After enabling the modem, it attempted to register on operator 732360 (ETB Colombia) instead of 732123 (Movistar). Manual registration timed out. Band lock was attempted for Movistar Colombia LTE Band 4 (1700MHz) and Band 28 (700MHz) via AT!BAND command, but registration remained idle.

Step 4 — QMI UIM Stack Failures

ModemManager repeatedly failed to initialize the modem despite SIM being readable via AT+CIMI:

```
QMI operation failed: Illegal SIM/USIM application state
state changed (locked -> failed)
failed reason: sim-missing
```

Step 5 — Recovery Attempts (Pre-Firmware Upgrade)

- AT+CFUN=0 / AT+CFUN=1 — SIM power cycle. NO IMSI persisted.
- AT!RESET — Full modem reset. Cleared failed QMI state temporarily but NO IMSI returned.
- mmcli -m 0 --reset — Same result.
- AT!BAND=00 — Reset to all bands. No change in registration.
- mmcli --set-allowed-modes=4g / 3g — Mode forcing. Registration stayed idle.
- mmcli --3gpp-register-in-operator=732123 — Manual Movistar registration. Timeout every time.

Step 6 — Firmware Upgrade

The proprietary fwdwl-litehostx86_64 tool failed with error 100 (Firehose protocol timeout). Firmware was successfully upgraded using qmi-firmware-update (libqmi-utils 1.30.4):

```
qmi-firmware-update -u -w /dev/cdc-wdm0 --device-open-qmi --ignore-mm-runtime-check \
SWI9X50C_01.14.02.00.cwe SWI9X50C_01.14.02.00_GENERIC_002.035_003.nvu
firmware update operation finished successfully
```

Step 7 — Post-Firmware Assessment

Parameter	Value	Assessment
Firmware	01.14.02.00	Successfully upgraded
AT+CSQ (signal)	25,99 (-83dBm)	Good signal
AT+CIMI (SIM IMSI)	732123667007142	SIM readable via AT
!GSTATUS MM state	NO IMSI	Modem stack not loading SIM
!GSTATUS PS state	Not attached	Cannot attach to network
Registration	idle	Never registers
Network scan	Timeout	Cannot scan operators

5. Key Evidence for Hardware Fault

- Signal strength is good (CSQ 25/31, WCDMA RSSI -64dBm) — coverage is not the issue.
- All other machines at the same location connect successfully to Movistar.
- SIM card is valid and readable via AT+CIMI — the SIM itself is not defective.
- NO IMSI persists across two firmware versions (01.08.04.00 and 01.14.02.00).
- NO IMSI persists across multiple modem resets (AT!RESET, AT+CFUN cycles, mmcli --reset).
- QMI UIM stack consistently reports 'Illegal SIM/USIM application state'.
- AT stack and QMI stack permanently out of sync — known symptom of failed SIM interface hardware.
- All software recovery options exhausted without resolution.

6. Note on Firmware Tool Failures

The proprietary Sierra Wireless firmware tool (fwdwl-litehostx86_64) failed consistently with error 100 (litefw_DownloadFW failed: 100) and error 103 (MBIM open response type error). Root cause was failure to complete the Firehose protocol handshake (sahara message timeout). Firmware was ultimately upgraded successfully using the open-source qmi-firmware-update tool from libqmi-utils.

7. Conclusion

The Sierra Wireless EM7565 modem on N096AV has a permanent hardware defect in its SIM interface controller. The defect manifests as the modem's QMI/UIM stack being unable to load the IMSI from the SIM card, preventing network registration. This fault persists across firmware versions, resets, and all known software recovery procedures. The modem must be physically replaced.

8. Recommendations

Priority	Action	Details
Immediate	Replace modem	Replace EM7565 on N096AV. Current modem is not recoverable.
Before deployment	Verify QMI mode	Confirm usb-devices shows qmi_wwan and mmcli -L detects modem.
Before deployment	Deploy modem-enable service	Install modem-enable.service to ensure MM enables modem at boot.
Before deployment	Upgrade firmware	Flash replacement modem to 01.14.02.00 using qmi-firmware-update.
Fleet-wide	Firmware audit	Check all machines for 01.08.04.00. Upgrade using qmi-firmware-update.